

1. Determine the truth value (True or False) of each proposition, applying definitions of logical connectives.

Proposition	Truth Value
a. $(y + y = 2y) \rightarrow (y - 8 > y)$. where y is a positive integer	False
b. $(32 = 4+5) \oplus (42 \text{ is divisible by } 7)$.	True
c. You have a quiz today unless it's not Thursday.	True
d. The square root of 1 is 1 if and only if $5+3 > 2*4$.	False
e. It is not the case that 15 is a multiple of 7.	True

2. Determine whether the proposition used Inclusive OR or Exclusive OR.

Proposition	Inclusive or Exclusive
a. When you buy more than 5 items you get a 20% discount or a free shirt.	Inclusive OR
b. Cash or credit.	Exclusive OR
c. To enter the country you need a passport or a voter registration card.	Inclusive OR
d. A password must have at least three digits or be at least eight characters long.	Inclusive OR
e. Experience with Java or Python is required for the job.	Inclusive OR

3. What is the negation of each of these propositions?

a.	Today is Thursday.
	Today is not Thursday
b.	Traffic is not heavy in EDSA.
	Traffic is heavy in EDSA
c.	$2 + 1 = 3$.
	$2 + 1 \neq 3$

4. Find the bitwise *OR*, bitwise *AND*, and bitwise *XOR* of the bit string 1001 1010 0111 and 0110 1101 1001.

bitwise <i>OR</i> ,	bitwise <i>AND</i> ,	bitwise <i>XOR</i>
1001 1010 0111 0110 1101 1001] Answer =1111 1111 1111	1001 1010 0111 0110 1101 1001 Answer = 0000 1000 0001	1001 1010 0111 0110 1101 1001 Answer = 1111 0111 1110